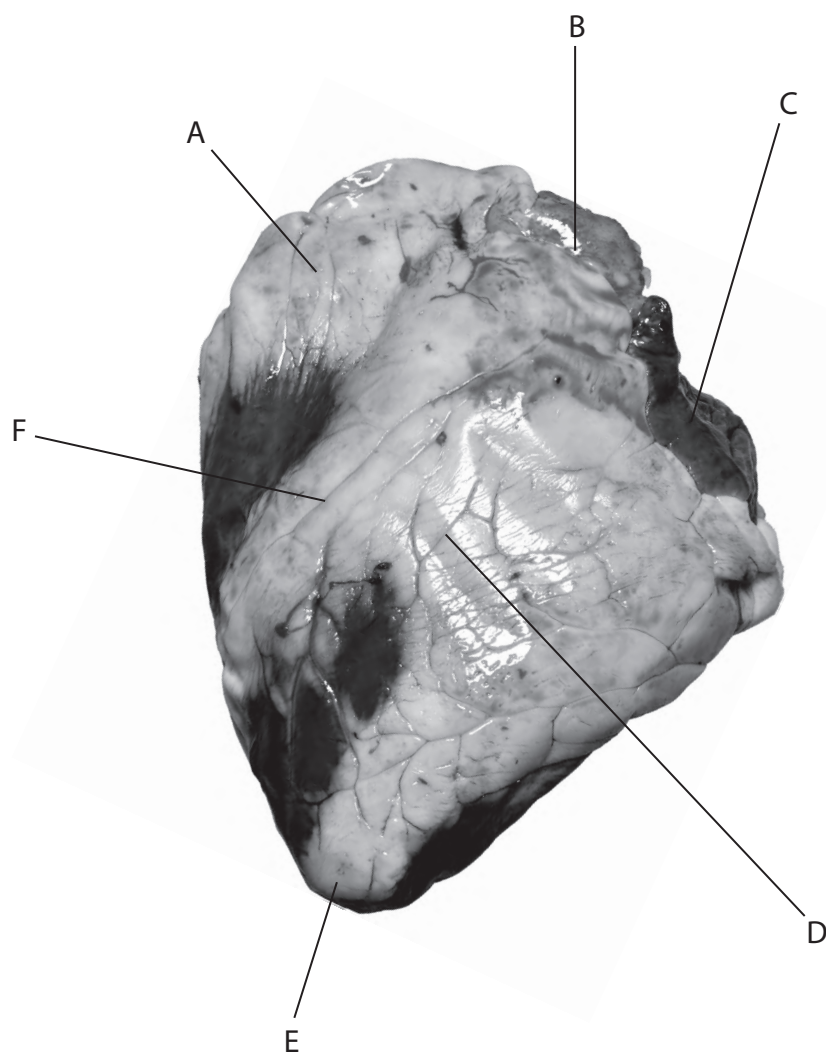


- 3 Photograph 1 shows the exterior of the front of the heart of a mammal. It has been positioned to begin a dissection.



(Source: © PjrStudio / Alamy Stock Photo)

Photograph 1

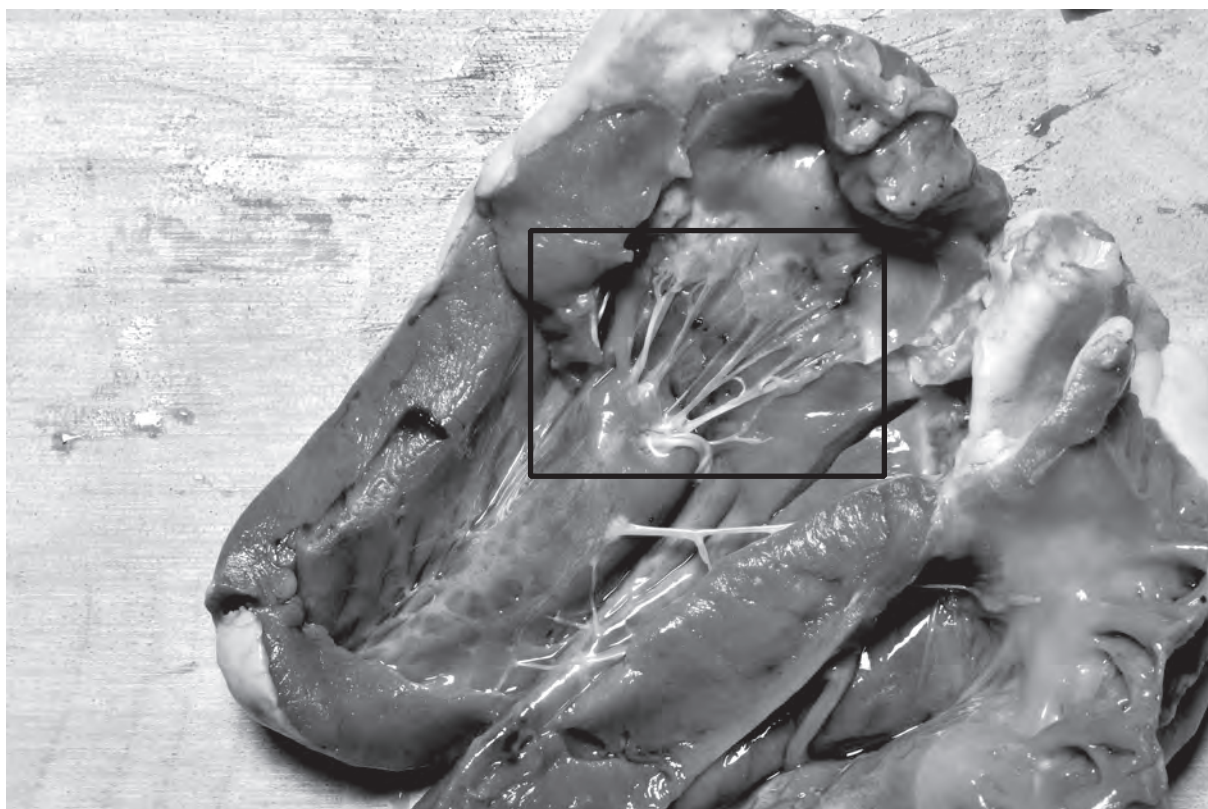
- (a) (i) Complete the table with the letters of the locations of the four structures listed.

(4)

Structure	Label letter
Apex	
An atrium	
The place where the right and left ventricles meet	
Coronary blood vessels	



- (ii) Photograph 2 shows the inside of the left side of the heart after it was cut open.



(Source: © PjrStudio / Alamy Stock Photo)

Photograph 2

Draw a line on **photograph 1** to show where this cut was made.

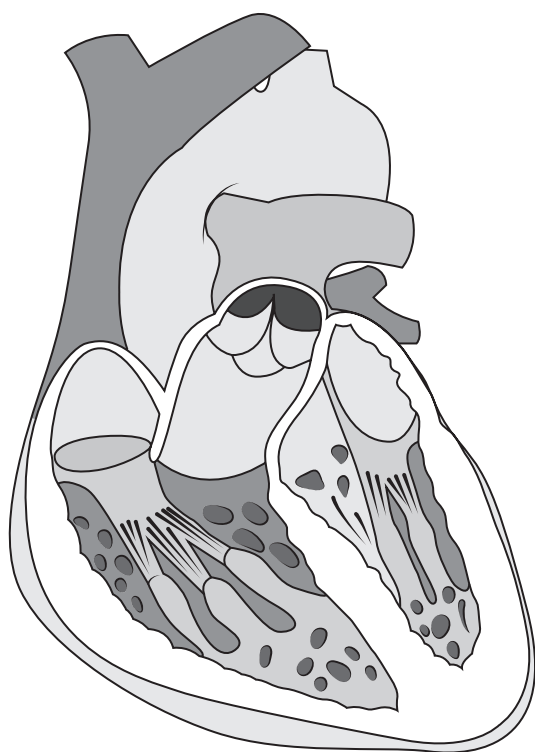
(1)

- (iii) Draw a plan of the structures within the rectangle on **photograph 2**.
Label the tendinous cords (heart strings) and the cut surface of the muscular wall of the ventricle.

(3)



- (b) The diagram shows a healthy heart and one showing a heart condition called dilated cardiomyopathy (DCM).



Healthy heart



Heart with DCM

Compare and contrast the structures of these two hearts.

(4)

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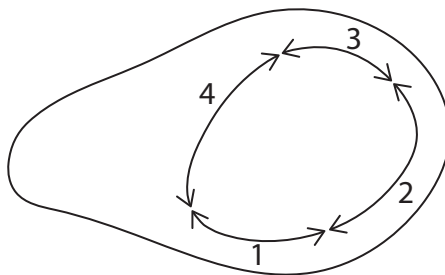
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- (c) A post-mortem study was made on 68 hearts. Of these, 48 were hearts with DCM and 20 were healthy hearts.

The diagram shows the measurements, labelled 1 to 4, that were made from sections of the left ventricles of hearts with DCM and those of healthy hearts.



- (i) State the independent variable in this study.

(1)

- (ii) When added together, these measurements give the full ventricular circumference.

The data is shown in the table.

Type of heart	Mean measurement / cm				
	1	2	3	4	Total
Healthy	1.88	3.20	3.26	4.92	13.26
With DCM	3.81	5.10	5.15	6.30	20.36

Calculate the percentage increase in the left ventricular circumference shown to be caused by DCM.

Give your answer to **three** significant figures.

(2)

Answer

- (iii) The mean circumference of hearts with DCM is larger than the mean circumference of healthy hearts.

Discuss the relative effects of the changes in the four individual measurements on the overall increase in circumference.

(3)

(Total for Question 3 = 18 marks)

TOTAL FOR PAPER = 50 MARKS

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